Question-4.3.6

EE25BTECH11022 - sankeerthan

Question

Passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.

Solution

let Passing through the points

$$\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix} \tag{1}$$

The direction vector of line passing through **A** and **B** is

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 1-3 \\ -1-4 \\ 6-(-7) \end{pmatrix} \tag{2}$$

$$= \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \tag{3}$$

The equation of line is

$$r\left(t\right) = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix} + t \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \tag{4}$$

graph

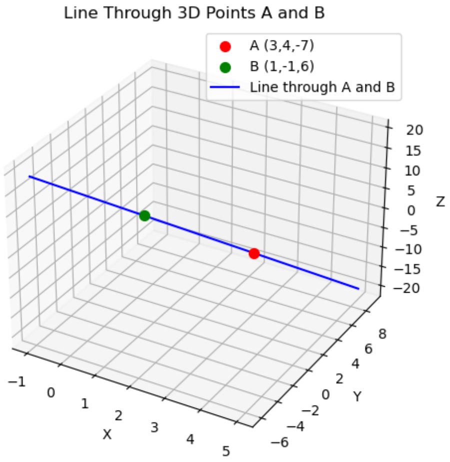

Figure:

C-Code

```c
// code.c
#include <stdio.h>

void get_points_and_line(double* A, double* B, double* n, double*
     k) {
    // Input points:
    A[0] = 3; A[1] = 4; A[2] = -7;
    B[0] = 1; B[1] = -1; B[2] = 6;
    // Precalculated normal vector and constant for these points
    n[0] = -12; n[1] = 23; n[2] = 7;
    *k = 7;
}
```

Python-Code

```
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

lib = ctypes.CDLL('./code.so')
```

Python-Code

```
def get_points_and_line():
    A = np.zeros(3, dtype=np.float64)
    B = np.zeros(3, dtype=np.float64)
    n = np.zeros(3, dtype=np.float64)
    k = ctypes.c_double()
    lib.get_points_and_line.argtypes = [
        ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
            c_double),
        ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
            c_double)
    ]
    lib.get_points_and_line(
        A.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        B.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        n.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        ctypes.byref(k)
    )
```

Python-Code

```
    return A, B, n, k.value

if __name__ == "__main__":
    A, B, n, k = get_points_and_line()
    print("Point A:", A)
    print("Point B:", B)
    print("Normal Vector n:", n)
    print("Constant k:", k)
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_points_and_line

# Extract from .so file
A, B, n, k = get_points_and_line()

# Direction vector of the line (B - A)
d = B - A

# Parameter range for the line
t_vals = np.linspace(-1, 2, 100)
line_points = np.array([A + t * d for t in t_vals])
# Plot
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

Python-Code

```
# Plot the two points
ax.scatter(A[0], A[1], A[2], color='red', label='A (3,4,-7)', s
    =50)
ax.scatter(B[0], B[1], B[2], color='green', label='B (1,-1,6)', s
    =50)

# Plot the line passing through A and B
ax.plot(line_points[:,0], line_points[:,1], line_points[:,2],
    color='blue', label='Line through A and B')

ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
ax.legend()
ax.set_title("Line Through 3D Points A and B")
plt.tight_layout()
plt.show()
```